

AMAR SINHA

Kolkata, India

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Summary

Computer Science (Data Science) undergraduate with hands-on experience in machine learning, computer vision, and full-stack software development. Skilled in Java, Python, JavaScript, and modern frameworks (React, Node.js), with applied project experience in YOLOv8 object detection, real-time multi-object tracking, and end-to-end web application deployment. Strong foundation in data structures, algorithms, and problem-solving, with a track record of building and shipping complete, production-deployed projects.

Education

Heritage Institute of Technology, Kolkata, India 2023 – 2027

B.Tech in Computer Science and Engineering (Data Science)

– CGPA: 1st – 7.15, 2nd – 7.95, 3rd – 8.87, 4th – 9.04, 5th – 8.66, 6th – 8.26

Methodist School, Dankuni, West Bengal, India 2021 – 2023

ISC (Class 12): 85 Percent ICSE (Class 10): 93 Percent

Technical Skills

Programming Languages: Java, Python, JavaScript, SQL, C, R

Frontend Development: React.js, Zustand, HTML5, CSS3, Tailwind CSS

Backend Development: Node.js, Express.js, Socket.IO, REST APIs

Databases: MongoDB, PostgreSQL

Machine Learning and Computer Vision: YOLOv8, DeepFace, ByteTrack, Scikit-learn, Machine Learning, Random Forest, Forecasting Models, Data Visualization, R Shiny

Developer Tools: Docker, Git, GitHub, Linux, VS Code, Postman

Deployment and Cloud: Render, Vercel, AWS EC2, AWS S3, Hugging Face

Experience

IIT Jammu 2026

AI Intern – Computer Vision and Deep Learning

Jammu, India

- Worked on object detection and multi-object tracking pipelines using YOLOv8, applying transfer learning and dataset fine-tuning to build models for real-time video analysis tasks.
- Certificate: [link](#)

Indian Oil Corporation Limited (IOCL) 2024

Intern – Comprehensive Study of Loyalty Program

Kolkata, India

- Conducted research on customer loyalty programs across industries and proposed data-driven strategies for improving customer acquisition and retention.
- Certificate: [link](#)

Projects

Golpo – Real-Time Chat and Video Calling Platform [GitHub](#) [Live Demo](#) *React, Node.js, Express, MongoDB, Socket.IO, WebRTC, Zustand*

- Developed and deployed a full-stack real-time communication platform supporting one-to-one messaging, image sharing, audio calls, and video calls using Socket.IO and WebRTC.
- Implemented Clerk authentication, real-time online/offline presence, persistent chat history with MongoDB, responsive UI, and global state management with Zustand; deployed the application on Vercel.

Football Match Analysis System [GitHub](#) *Python, YOLOv8, OpenCV, Supervision (ByteTrack), Scikit-learn, Streamlit*

- Built an end-to-end computer vision pipeline to detect, track, and analyze players, referees, and the ball in football broadcast footage using a custom-trained YOLOv8 model fine-tuned on a Roboflow dataset, with ByteTrack for persistent multi-object tracking across frames.
- Implemented automatic team classification via K-Means clustering on jersey colors and a proximity-based ball-possession algorithm to compute live team ball-control statistics.
- Deployed an interactive Streamlit web application allowing users to upload football clips and receive an annotated output video with live tracking, team colors, and possession statistics.